

Season 2 — Building Something That Can Say No

Research document. Thirteen episodes. Season 1 laid out a theory. This season turns four sentences of it into code, and spends half its runtime on the part nobody teaches: **getting the text right before you count anything.**

*Two anchors per concept from here on. A **character** for the audience, and a **clinical mechanism** for you, because that is how you actually hold a concept and the audience does not have it. Use the character on camera. Use the clinical one to know you have understood it.*

What this season is doing

Season 1 ended with four word-lists that do not exist yet. This season builds them, and the build has three separable problems that get confused constantly.

#	THE PROBLEM	THE QUESTION	EPISODES
1	the charge	why does any of this need to be measurable at all	1-3
2	the corpus	which texts, and is the one on my disk the one I think it is	4-7
3	the lexicon	does this word mean what I am counting it as	8-10
4	the metric	what number do I compute, and what does it throw away	11-13

Which part is load-bearing

Problem 2 is the one that decides everything and gets zero attention anywhere.

Everyone argues about problem 4, the statistics. Nobody checks problem 2. And in this project, problem 2 produced **two separate bugs**, both of which shipped plausible output instead of errors, and one of which silently deleted an entire control group from a study.

A statistic computed on the wrong text is not a weak result. It is not a result. That is the sentence carrying this season.

What is not decided yet at the end of this season

Nothing has been tested. Twelve episodes in, there is an instrument and no result. That is deliberate and it should be said in episode 1, because the temptation in every methods project is to look at an answer while still building the thing that produces it.

Collecting Season 1

Eighteen episodes gave you a theory: copied desire, the model becoming the rival, rivals converging, a crisis of distinctions, a community discharging it onto one victim, and four marks left behind in the texts.

Season 1 episode 14 gave you five objections to it. **This season is about the one that has an answer**, and about what answering it costs.

The character

Sherlock.

He looks at a man's cuff and tells you where he has been, and he is right, and it is the best thing in the episode. Now ask what makes him right. **The writer composed the deduction first and then arranged a world in which it holds.** No scene was ever filmed in which Sherlock reads the cuff and the man turns out to have been somewhere else.

The method is never wrong because nothing was ever allowed to go wrong, and the audience experiences that as brilliance, because from the inside there is no difference between a method that always works and a method that is never tested.

Sharper, and without the fiction: a cold reading. *"You have been hurt by someone close to you."* It cannot fail. Not because it is true, but because **there is no state of the world in which the reader is caught out.**

That is the shape to hold on to. **Neither one is a lie.** Both are impressive. Both are constructed so that no possible observation embarrasses them.

The clinical anchor

A test with 100% sensitivity and zero specificity. It comes back positive in every patient. It is never wrong.

Follow what that does to a clinic. A positive result changes nothing, because you would have got one anyway. A negative result never happens, so it cannot reassure anybody. **You would make exactly the same decisions if the machine were unplugged**, and the only thing the test reliably produces is the feeling of having checked.

It is worthless, and for exactly the reason above: **a test that cannot come back negative carries no information.**

Write that on the board and leave it up for thirteen episodes. **The entire season is one long attempt to build a test that can come back negative.**

And notice that the fault is not in the test's accuracy. It is right every single time. **Being right about every case is compatible with telling you nothing**, and that sentence is the whole reason this project exists.

The device: build the confirmation table

Do not assert that Girardian reading is unfalsifiable. **Demonstrate it**, by taking every possible observation and asking what a Girardian says about it.

WHAT YOU FIND IN A TEXT	WHAT A GIRARDIAN READING SAYS
the victim is guilty	that is the persecutors' account, exactly as predicted
the victim is innocent	that is the revelation breaking through
there is a crowd	the mob, exactly as predicted
there is no crowd	the mechanism has been concealed, which is what myth does
the violence is described	the residue is visible
there is no violence	the sacrifice has been ritualised and hidden
the text blames the victim	persecution text
the text pities the victim	partial revelation, tragedy sits here

Go down the rows in pairs, because the pairing is the trick. **A crowd is evidence**. No crowd is also evidence, because concealing the mechanism is what myth is for. **Described violence is evidence**. Absent violence is also evidence, because ritual hides it. **A guilty victim is evidence**, since that is the persecutors' version. An innocent one is evidence too, since that is the revelation getting through.

Read the right-hand column down. There is no row where the theory loses.

And be careful about the objection you are making, because the easy version of it is wrong. **Every single row is defensible on its own**. Each is a reading a serious person could hold, argued from Girard's actual text, and some of them are excellent. **Nothing is refuted here at the level of the row**.

The load-bearing sentence is about the column. A theory that has an answer for each observation individually, and an answer for the negation of each observation, has told you in advance that it will survive whatever you find. **You learn its verdict before you open the book**, and a verdict available before the evidence is not a verdict.

That is the charge, and it is not a strawman. It is what serious critics say, and the table is the reason it stings.

And the defence people usually give makes it worse

Somebody objects, and the Girardian responds with **another reading**. A better one, often a beautiful one, and frequently one that does genuinely illuminate the text under discussion.

Another reading is another positive result from a test that has never returned a negative. Adding a row to the table does not fix the table.

Worse, it strengthens the objection while feeling like an answer. **Each new reading demonstrates the theory's reach**, and reach is precisely what is not in question. Nobody doubts that Girardian reading can be applied to anything. That is the complaint.

What the only real answer looks like

State what the theory forbids, in advance, in a form somebody else can check.

Not "here is a text that fits." Anyone can produce that. Instead:

- **here are the texts I say will score high**
- **here are the texts I say will score low**
- **I wrote it down before I looked**
- **and if the low ones beat the high ones, I was wrong**

Every line there is doing separate work. The first two make the theory commit to a partition of the world rather than to an interpretation of a passage. The third removes the author's memory from the evidence, which matters because **nobody knows what they expected once they have seen the answer**, and honesty does not help with that. The fourth is what makes the whole arrangement a test rather than a display, because it names the outcome the author is agreeing to lose on.

That last line is the entire design of the next six seasons. **The instrument exists to make a row in that table come out empty.**

The honest cost, and it is stated in episode 1 rather than discovered later

Making the theory testable makes it smaller.

The instrument can only reach the parts of Girard that leave countable residue, which is Season 1 episodes 6 to 10 and nothing before them.

Season 1 episodes 1 to 5 are the most interesting part of Girard and nothing in this project tests any of it. Say that here rather than letting a critic say it in a comment.

And there is a second cost, which is subtler and worth naming.

A testable version of a claim is not the same claim. When you convert "no dissenting voice is ever heard" into "a word-list fires at a higher rate in one tier than another," you have built something **adjacent** to Girard's sentence, not a translation of it.

Look at what slid. Girard's sentence is about **a voice**, a person speaking at a particular moment, and it is absolute: never, not once, nowhere. The operational version is about **a rate**, which is an average over a whole text, and it is comparative: more here than there. **Neither the moment nor the absoluteness survived the conversion**, and both of them were the claim.

And the danger is not that the conversion is careless. It is that the result still passes as Girard's sentence in conversation, so a passing score gets reported as support for something the test never touched.

Every season after this one is haunted by that gap, and **Season 7 episode 4 is where the gap finally does visible damage.**

What would count against this episode's own claim

Because a season about falsifiability should apply it to itself:

1. **A Girardian reading that names, in advance, a text it expects to fail** and then reports the failure. If that exists in the literature, the confirmation table above is unfair.

2. **A row in the table where the theory actually loses.** Find one.
3. **The instrument turning out to confirm everything too.** If the numbers also never come back negative, this project has reproduced the disease with arithmetic. **Watch for that. Season 3 is where it would show.**

Handing off

The answer is to write down what you expect **before** you look.

Episode 2 is the moment that happens: four word-lists, sixty-six words, written before a single text had been downloaded.

Questions

1. **Name a claim you hold that could not come out negative.** Not a bad one. One you actually believe.

2. Add a row to the confirmation table from your own field. **What observation would embarrass a theory you like?**

3. Is "unfalsifiable" a fatal objection to a theory of **literature**, or only to a theory of **nature**?

4. If making a theory testable shrinks it, **when is that trade worth taking?**

Collecting episode 1

Episode 1's confirmation table had no losing row. The only escape was to **write down what you expect before you look**.

This episode is that moment.

The character

Ippo announcing the Dempsey Roll. He tells you what is coming. Then it lands or it does not, in front of everybody, and **there is no version where he explains afterwards that something else was the plan**.

The famous one: Babe Ruth's called shot. People still argue about it a century later for one reason: **calling the shot is the only thing that converts a home run from an outcome into a claim**.

The clinical anchor

A pre-specified primary endpoint. A trial that decides after unblinding which outcome it was measuring has not measured anything. Everybody in medicine knows this. **It is the same idea exactly.**

And notice what the rule is actually protecting against, because it is not fraud. A trial with twenty outcomes and no primary will find something significant by chance, and the investigator who reports it is not lying. **He is remembering, sincerely, that this was the one he cared about**, and there is no way for him to check that memory against anything. The endpoint is fixed in advance because nobody's recollection of their own expectations survives seeing the answer.

What was actually built

`make_scorers.py`. **Four word-buckets, 66 words total, written before a single text had been downloaded.**

One bucket per stereotype from Season 1 episodes 6 to 9, which is the whole design. **The theory says persecution leaves four kinds of trace, so the instrument has four containers**, and each one holds the English words that would carry that trace into a text.

The ordering is checkable in the repository: the corpus arrives in a later commit than the lexicon.

BUCKET FIRST DRAFT

crisis	plague, pestilence, famine, drought, blight, barren, sterile, sickness, disease, death, twin, twins, double, doubles, likeness
crimes	incest, parricide, patricide, regicide, murder, betrayal, poison, poisoning, sacrilege, blasphemy, defilement, pollution, abomination, unclean, monstrous
marks	stranger, foreigner, alien, cripple, lame, blind, deformed, beggar, orphan, king, queen, tyrant, monster, scapegoat, accursed, marked
violence	crowd, mob, unanimous, stone, stoned, cast, expel, banish, exile, drive, driven, slay, slain, kill, blood, hunt, hound

The device: the dead words are the receipt

Here is the thing nobody notices, and it is the strongest evidence in the whole project that the ordering was genuine.

Nine of those 66 words never fire at all, in roughly two hundred thousand words of Greek and Roman literature:

sterile · patricide · regicide · blasphemy · poisoning · cripple · orphan
· scapegoat · stoned

Now ask the question that turns this from a technical footnote into evidence:

What would this list look like if the lexicon had been written AFTER reading the texts?

It would have no dead words in it. Nobody sits down with the corpus open, notices that regicide never appears, and puts it in anyway. You would have removed them, silently, without a moment's guilt, because removing a word that does nothing is housekeeping. **The bucket would look tidier and better**, and nothing anywhere would record that it had been tidied.

A lexicon with nine dead words in it is a lexicon that was written blind. The failures are the fingerprint.

Understand what kind of argument that is, because it is not the usual kind. It is not a claim about the author's character and it does not ask anybody to believe him. **It points at a property of the artifact that would have been destroyed by the behaviour being suspected.** Tidying leaves no trace; the absence of tidying does. The nine words are evidence in the way an unpicked lock is evidence.

And it survives the obvious counter-move, which is worth stating rather than hoping nobody thinks of it. Somebody could plant dead words deliberately, to manufacture exactly this receipt. True, and to do it he would have to have anticipated this argument before writing the lexicon, and then commit to carrying useless words through every later analysis where they cost him coverage. **The forgery is possible and it is not free**, which is the most any evidence of this kind ever offers.

That is a much better argument than a commit timestamp, and it is available to anyone who reads the file. **Season 2 episode 3 is about why the timestamp on its own is weaker than people think.**

And the fourth dead word is a finding, not a footnote

Read the list again and stop on scapegoat .

The word does not appear anywhere in the corpus. **The instrument built to detect scapegoating cannot find the word "scapegoat,"** because that is not how these texts talk.

And it goes further. parricide, patricide and regicide are the Latinate labels for exactly the crimes Girard's theory is about, and **all three fire zero times.**

The theory names its objects with nouns. The literature describes them with verbs.

"He slew his father." Not "he committed parricide."

And that is a difference in what the language is doing, not a difference in spelling. **A noun is a category with the event already classified inside it.** Parricide is a verdict; somebody has decided what the killing was and filed it. The verb reports the killing and leaves the classification to the reader, which is precisely what a narrative does and precisely what a theory does not.

So the instrument was built out of the vocabulary of judgement and pointed at texts written in the vocabulary of event.

That gap is not a technicality about word choice. **It is a gap between the vocabulary of a twentieth-century theorist and the vocabulary of the material he is theorising about,** and repairing it is most of episode 9.

What would count against this

1. **A dead word that is actually a coding error.** If `stoned` fails to fire because of a tokenisation bug rather than because the texts do not use it, the receipt is worthless. **This is a real risk and episode 6 is about exactly that class of bug.**
2. **The lexicon having been quietly edited after the fact.** The defence is not trust, it is the file history, and episode 3 examines how good that defence actually is.
3. **Sixty-six words being too few to measure anything.** A real objection. The repaired version has several hundred surface forms and episode 9 explains why that is not simply an improvement.

Handing off

The lexicon was written first. **Episode 3 asks what "first" is worth,** and the answer is less than you would like and more than it looks.

Questions

1. Writing the lexicon before the corpus protects against **one specific failure.** Name it precisely.

2. **The dead words are the receipt.** Is that argument sound? What would make it unsound?

3. Nine dead words out of 66. **Bad lexicon, or an interesting finding about how texts talk?**

4. **What would you have put in the buckets?** Write your list now, before episode 9 shows you the repaired one.

Collecting episode 2

Episode 2 found evidence the lexicon was written blind: **nine dead words that nobody would have left in.**

But that is an argument from the *contents* of a file. The project's own claim is about **order**, that the predictions were written before the scoring, and it offers a different kind of proof for that. **This episode is about how good that proof is, and the answer is uncomfortable.**

The character

A sealed envelope handed to a lawyer.

The trope exists in a hundred films for one reason: **it converts a claim made afterwards into a claim made before**, and every audience understands instantly why that matters. Nobody needs the concept explained. The whole of pre-registration is in a piece of stationery.

And then the version that shows the crack, which is Borden's diary in The Prestige. A rival steals it, reads it, and reasons carefully from it, because a diary is a dated record and a dated record is evidence. It was written to be stolen. The dates are real, the entries are in order, the object is exactly what it appears to be, and the reader is still being played, because the one thing the artifact cannot record is the intention of the man holding the pen.

Hold both. **The envelope is why the procedure works. The diary is why this episode exists.**

The clinical anchor

A trial registration number. Not the protocol, the **registration**, which is the thing whose entire value is that it was filed on a date and cannot be edited.

And notice what a registry is really doing, because it is not storage. The protocol could sit in a drawer. **The registry's function is that it is held by somebody with no stake in the result**, so the record is outside the reach of the person the record is about.

The rule the project follows

1. Write the predictions in a file.
2. **Commit it to git.**
3. Then run the scorer.
4. Report what happened, failures at the top.

And the reason, in the project's own words, which is the sentence to read aloud:

"If the rule is written afterward, it gets shaped by what was found, and **the shaping is invisible from the inside**. I would have written a definition that produced the answer I already liked **and experienced that as being careful.**"

That is the claim. Not that people lie. That **a mind adjusting a rule to fit a result cannot feel itself doing it**, so the only defence is a record somebody else can check.

And now go and check it

This is where the episode turns.

ROUND	PRE-REGISTRATION	RESULTS	GAP
1	21 Jul 01:37	21 Jul 01:39	2 minutes
2	21 Jul 03:00	21 Jul 03:14	14 minutes
3	22 Jul 00:55	22 Jul 03:22	2 h 27
NT count	24 Jul 22:53	24 Jul 23:09	16 minutes
Layer 2	25 Jul 00:56	25 Jul 00:58	2 minutes
4	11 Aug 02:07	11 Aug 02:17	10 minutes

Median gap: fourteen minutes. Every prediction and its result were committed inside one continuous working session, often minutes apart.

Two minutes, twice. Sit with that number before defending it.

The defensive reading is available and it is genuinely true. The work was done in long uninterrupted sittings. The scorer runs in seconds. And there is no reason a careful person would pause between freezing a prediction and executing it. **Everything about the gap is consistent with the procedure having been followed exactly.**

It is also consistent with the opposite.

The hostile reading, stated at full strength: you had the numbers on screen and committed the prediction file first. **Git proves which of two files hit the disk first. It does not prove that no scoring happened before either.**

That is the whole weakness in one line. A commit records **when a file was written down**, and the claim being made is about **when something was known**. Those come apart, and nothing in a timestamp closes the distance. This is Borden's diary again: an accurate, dated, unfalsified record that answers a question adjacent to the one being asked.

That objection is real and it is not answered anywhere in the project. Say so out loud, before anybody else gets to.

The device: stop looking at WHEN and look at WHAT IS INSIDE

Here is the fix, and it is better than the thing it replaces.

Do not ask when the commit was made. **Ask what files it contains.** A commit is not a timestamp with a note attached. It is a complete snapshot of the project at a moment, so it records not only what was said but **what existed**.

COMMIT	CONTAINS	DID THE SCORER EXIST YET?
pre-registration 3	the prediction file, and nothing else	no
results 3	results, and nulltest.py , and the output	it arrives here
pre-registration 4	predictions, lexicon, tiers	no scorer
results 4	results, and score_dissent.py	it arrives here

For those rounds, git proves something **much stronger than the order of two prose files:** the **program that produced the number did not exist** when the prediction was frozen.

You cannot fit a result to a prediction using software you have not written yet.

Notice why that argument is stronger, and it is not because it is more rigorous in style. **The timestamp defence asks you to accept a claim about a person's behaviour**, which is unverifiable in principle. The manifest defence rests on an absence in the artifact: the file is not there, and its absence is checkable by anybody, forever, with no cooperation from the author. **A defence that depends on what somebody did is replaced by a defence that depends on what a repository contains.**

That is the argument to make on camera. Not the timestamp. The manifest.

And the exception, which is the part that costs something

Round 2's pre-registration commit contains the scorer and two files of already-scored results.

They are scores on a *different* corpus, which the pre-registration itself discloses as a smoke test, and that disclosure is real and should be reported. But apply the manifest test as stated and it fails: **the machinery was fully operational and committed at the moment the predictions were frozen.** The gap to results is fourteen minutes. Every protection this episode just built is unavailable for that round.

And round 2 is the round that produced the project's only surviving positive number.

The least protected round is the one carrying the result.

Say that plainly. It is the single most damaging fact about the project's evidence and it is better said by you. **And say it in this position**, in the middle of the episode that constructs the defence, rather than in a footnote after the audience has been persuaded.

What would count against this

1. **A round where the scorer appears in the pre-registration commit and the result is claimed anyway.** That is round 2 and it is disclosed above.
2. **Files edited after commit without a record.** Git makes that visible; the defence is that anyone can check.
3. **The whole file-manifest argument failing** if someone wrote the scorer, deleted it, committed the prediction, and restored it. **Possible. Undetectable. And at that point you are not doing science, you are doing fraud**, which no procedure defends against.

Handing off

Predictions are frozen and the order is as good as it is going to get.

Episode 4 asks the question that comes before all of it and almost nobody asks: which texts, and why those?

Questions

1. **Is the file-manifest defence enough?** What procedure would satisfy a reader who trusts nothing at all?
2. Two minutes. **Does that bother you more or less than you expected** before you saw the manifests?
3. Design a pre-registration procedure that **does not depend on trusting commit order**.
4. The least protected round carries the only result. **How much should that discount it?** Put a number on it.

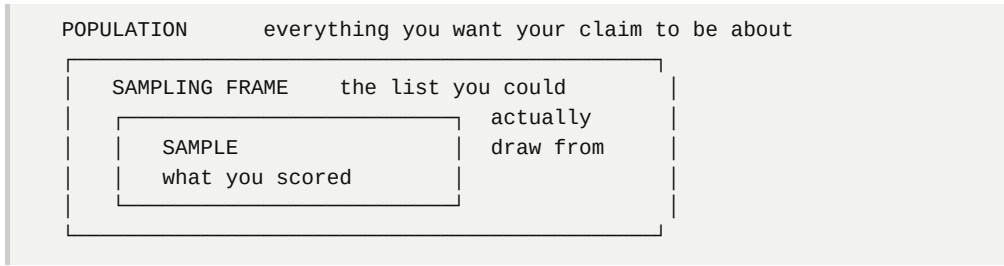
Collecting episode 3

Episodes 2 and 3 secured **the instrument**: written blind, frozen, order checkable.

None of that helps if you point it at the wrong texts, and almost nobody asks how the texts were chosen. This episode is that question.

The device: three circles, and the gap between them

Draw these on camera and keep them up.



IN THIS PROJECT	
population	all Greek and Roman narrative literature
sampling frame	what two websites and Project Gutenberg happen to hold, in English translation
sample	76 texts, later 98

Every claim you make is about the population. Every number you have is about the sample. The frame is the gap between them, **and it is almost never examined.**

Sampling gets discussed constantly and it is nearly always the inner boundary being discussed: was the sample big enough, was it drawn at random, is it representative of the frame. **All of that concerns the smaller circle.** The outer boundary, between the frame and the population, is the one that decides whether the question was ever answerable, and it does not appear in the statistics at all, because **no procedure inside the frame can detect the shape of the frame.**

The clinical anchor, exact

Eligibility criteria against the patients who actually walk through your door.

The paper says "adults with heart failure." **The population** is adults with heart failure. **The frame** is adults with heart failure who attend a tertiary centre, speak the language, and consent. **The sample** is who enrolled.

Everyone knows the drug behaves differently outside the frame. **The abstract says none of this.** And the reason it does not is worth naming without cynicism: the frame is not a decision anybody made. It is the accumulated residue of where the building is, who can get to it, who was referred, and who said yes. **Nobody chose it, so nobody feels responsible for reporting it.**

The character anchor

Truman.

Everything he believes about the world is correct about Seahaven. He is not foolish, he is not incurious, and his reasoning from evidence is sound. **The frame was built around him and it has no visible edge**, so every generalisation he forms is confirmed by everything he can reach.

Notice what would not have helped him. More observations, better records, a larger notebook, more rigour. All of it takes place inside the frame and all of it comes back confirming. The only move that works is the one that ends the show, which is sailing at the wall to find out where it is.

A corpus is Seahaven. Every statistic computed on it is honest, and it describes what somebody digitised.

What this project's frame actually was, biases stated out loud

THE FRAME IS BIASED TOWARD	CONSEQUENCE
texts old enough to be out of copyright	translations are 1900 to 1921, not modern
texts somebody bothered to digitise	the canonical and the famous
texts with an English translation at all	fragments and minor works vanish
whatever one website's page structure let a script fetch	this produced two silent bugs

Dwell on the last row. The sampling frame was partly determined by **HTML**. Not by scholarship. Not by a research design. **By how a website laid out its pages, and by whether a script followed a link.**

Who is actually in this corpus, since the names run through six seasons

Before the numbers, put the cast on screen once. **These names recur in every season from here** and nothing later stops to say who they are.

NAME	WHEN	WHAT IT IS
Homer	c. 8th c. BC	the <i>Iliad</i> and the <i>Odyssey</i> . Greek epic verse
Euripides , Sophocles, Aeschylus	5th c. BC	Athenian tragedy, written for the stage
Seneca	c. AD 50	Roman tragedy in Latin, on the same Greek stories
Ovid	AD 8	the <i>Metamorphoses</i> , a Latin verse poem retelling myth as a chain of transformations
Apollodorus	1st-2nd c. AD	the <i>Bibliotheca</i> , a bare prose catalogue of the whole of Greek myth
Hyginus	c. 2nd c. AD	the <i>Fabulae</i> , the same idea as Apollodorus and even more compressed
Diodorus Siculus	1st c. BC	a universal history whose early books retell myth as history
Statius , Quintus Smyrnaeus, Apollonius	1st-4th c. AD	later epic, filling gaps the earlier poems left

Two distinctions inside that list matter for everything downstream.

Verse against prose. Homer, Euripides and Ovid are poets writing to be performed or read for pleasure, with description, speeches and imagery. Apollodorus and Hyginus are writing reference works, with none of that.

That split is the one the project calls epic against mythography, and it is about to become one of the most load-bearing facts in the study.

Here is why. The instrument counts hits per thousand words. So it treats a page of Hyginus and a page of Homer as the same kind of object. **They are not remotely the same kind of object.** One is a summary with the description stripped out. The other is a poem, and most of its words are description.

The number that proves the frame is not neutral

The corpus contains **36 epic texts** and **8 mythography texts**.

That is not a statement about Greek literature. **It is a statement about how the Iliad and the Odyssey were split into books by whoever digitised them, and how Apollodorus was not.**

Follow the mechanism, because it is duller and worse than a bias. Two epics arrived as one file per book, so two works became dozens of rows. One mythographer arrived as a handful of files, so a comparably large body of material became a handful of rows. **Nobody weighted anything. Nobody decided epic deserved more say.** A genre's influence over every average in the project was set by an upload convention.

A genre's weight in the corpus is an artifact of file-splitting, and every genre mean in Season 3 inherits it. When mythography ties tragedy in Season 3 episode 7, remember that mythography is eight files and epic is thirty-six for reasons that have nothing to do with antiquity.

And note which direction the exposure runs. **The instrument's precision is unaffected**, so nothing about this shows up as noise or as a wide interval. The numbers stay just as tight and are computed over a differently shaped world.

The number that has four values, and this one is checkable live

Here is something the project never printed, and it is the best exercise in this episode because the audience can verify it while watching.

Ask how many texts are in the corpus. The repository answers four different ways, and three of the four are in the same results file.

FIGURE	WHERE IT APPEARS
76	every file in <code>corpus2/</code> , and every row in the results table
73	never stated, and it is the one that is derivable
71	the pre-registration and the results file, called "scoreable"
72	the same results file, for the <i>Bacchae</i> 's rank and the saturation count

Now derive the 73. The pre-registration declared a minimum text length of 800 words. Three texts in the table fall under it, at 613, 689 and 777 words. Take them out of the 76 and you get **73**, exactly, and that is the number of texts that the project's own declared rule permits.

So the figure the rule produces is the one figure nobody ever wrote down, and two other numbers are used instead, in the same document, without either being reconciled to it.

Be careful about what that does and does not show. **Nothing here is a falsified result.** Nobody has been caught. The gap between 71 and 73 is two texts and nobody knows which two, and the effect of any of this on the published findings is almost certainly nothing.

What it shows is that the sample was never counted. The corpus was assembled, a rule was written about it, and the rule and the pile were never once put next to each other. Which is the identical failure to the missing integrity check from episode 5, at the level of the sample instead of the file, and it is why the rule in episode 7 has to be mechanical rather than remembered.

The one thing done right, and it is worth the episode

The control group for one study was taken from **Project Gutenberg's own "Top 100 last 30 days" list**, applied mechanically **in the order the list gives them**.

The rule was fixed in advance: single-author prose fiction over 40,000 words, nothing by an author already in the test set, no anthologies, no plays, no poetry.

"The ordering is not mine." That sentence is the whole design. A list generated by strangers downloading books, applied in its own order, **cannot be shaped by what I wanted to find**.

Both halves of that are load-bearing and the second is the one people drop. Choosing an external list is easy. **Taking it in its own order is what removes the discretion**, because a list you are allowed to walk down and pick from is your list again, with a citation attached. The rule fixed in advance does the same job for the exclusions: it decides what counts as eligible before anybody knows which books are affected.

Compare that with choosing forty myths yourself, which is Season 1 episode 14's objection 1, and which is the charge this entire project exists to answer.

What would count against this

1. **The genre imbalance changing the results.** If reweighting epic to eight files moved the genre means, every Season 3 conclusion is about file-splitting. **Nobody has ever run that check.**
2. **The four corpus sizes having an innocent explanation each.** Likely, and it would not help. **A figure that needs an explanation per usage is not a figure the reader can check**, and the point of the episode is checkability rather than correctness.
2. **The translation window mattering.** Everything is 1900 to 1921 English. Season 3 episode 5 tests this once, on one paired play, and never again.
3. **The frame excluding the texts that would falsify.** If the mechanism shows up mainly in material nobody digitised, the corpus cannot see it, and no amount of statistics repairs that.

Handing off

You have chosen the texts. **Episode 5 asks whether the text on your disk is the text you think it is**, and the answer, twice, was no.

Questions

1. **Write your own frame for "Greek literature."** What does it exclude, and does the exclusion favour any conclusion?
2. Epic is 36 of 76 texts because of file-splitting. **How would you fix that without choosing an answer?**
3. An externally generated list applied in its own order. **What is the strongest objection to that as a control?**
4. Which of the three circles do you think is doing the most damage here?

Collecting episode 4

You have chosen which texts belong. **This episode asks whether the file on your disk is the text you chose**, and in this project the answer was no. Twice.

The clinical anchor, and use this one, it is the whole episode

A haemolysed sample.

The potassium comes back at 6.8. The number is real. The machine is calibrated. The lab is competent. **And the patient does not have hyperkalaemia.**

Nothing about the analysis is wrong. **The specimen was wrong.** And the result looks completely normal for a sick patient, **which is why people act on it.**

That is the shape of every corpus bug in this project, and the shape is what makes them survivable.

The character anchor

The cat in The Matrix.

A black cat walks past a doorway. A moment later the same cat walks past the same doorway the same way. **Everyone in the room registers it and nobody stops**, because that experience has a name and the name is déjà vu. It arrives filed under "ordinary quirk" before anybody has examined it.

It was a glitch, and it was **the only glitch anyone got**, and the reason it went past is not that it was subtle. It is that the world had already supplied a category that absorbed it.

That is what corpus bugs look like on screen. Not an alarm. A slightly odd thing that resembles the odd things you have taught yourself to ignore.

The device: a bug that errors versus a bug that publishes

Draw the distinction and keep it up.

	WHAT HAPPENS	DO YOU NOTICE?
loud failure	the script crashes, the file is empty, the number is obviously absurd	yes, immediately
quiet failure	the script completes, the file has content, the number is unremarkable	no. It goes into the results

Every bug that matters is the second kind, because the first kind is fixed in the same afternoon it is created. **A bug that produces plausible output is not a bug you find. It is a bug you publish.**

And there is a selection effect underneath that which is worth stating as a general law rather than as a remark about this project. **Your attention is allocated by how loudly things fail.** Loud failures are removed on the day they appear, so the population of surviving defects is not a random sample of the defects you made. **It has been filtered**

for silence, continuously, by your own competence. The better you are at fixing what shouts, the more purely the remainder consists of things that do not.

Bug one: the play with its ending missing

get_corpus2.py downloaded pages from a Greek text archive. The archive splits long pages and links the remainder as `CONTINUED`. **The fetcher never followed the link.**

TEXT	AS SCORED	ACTUAL	MISSING
Seneca, <i>Hercules Oetaeus</i>	11,119 words	20,621	46%

And the missing half is not random. The file broke off after one speech, so Deianira's death, Hercules on the pyre and the apotheosis were **all absent**. Hercules spoke exactly one line in the file. Two characters never appeared at all.

The scorer was reading a play with its sacrifice removed. And the score came out mid-table, and **nothing flagged it.**

Stay on the coincidence for a second, because it is not really one. The instrument exists to detect the residue of a sacrificial killing. **The half of the file that went missing is the half containing the killing.** That is not bad luck: the truncation happened at a page break partway through, and in a tragedy the death comes late, so **any truncation of any tragedy removes the ending, and the ending is where the instrument's entire subject lives.** A bug of this shape will damage this project's measurements more than almost any other kind, and it presents as a mid-table score.

Mid-table is the worst possible output. A zero would have been investigated. An implausibly high score would have been investigated. **A perfectly ordinary number is the one result nobody asks a question about.**

And it was nearly missed twice more

The diagnostic that eventually caught it classified the string `CONTINUED` as a **one-off speaker name** and dropped it into a list of ignored oddities.

That marker was in the very first line of diagnostic output, and it was read past twice.

Say that out loud. The evidence was on screen, in the first line, and it did not register, **because it looked like the kind of noise you learn to ignore.**

Bug two, nineteen days later, same shape

A second study's control-group fetcher scraped a top-100 list by anchoring on a string in the page. The site later began using that same string **first in a navigation link**, so the scraper anchored 35,000 bytes too early and its stop condition truncated the chunk **204 characters later**, before any list item.

```
old anchor -> 0 list items, silently empty control group
new anchor -> 100 list items
```

The study's own results file recorded the cause as "the fetch was killed partway through." That was the symptom. **The study ran for months without a control group,**

and the reason was one string match.

Notice that the misdiagnosis is itself the pattern repeating. A killed fetch is a loud, ordinary, uninteresting explanation, and it fits the evidence, and it is wrong. **The author reached for the noisy cause because a quiet cause does not present itself as a cause at all.** The site had changed underneath the script, which is the class of failure that never announces itself, because nothing on either side of it broke.

The recommendation nobody implemented

From the correction file, written in July after the first bug:

"No integrity check runs over the corpus. This defect was found by accident, on the way to something else. **A truncated text is invisible to every metric currently computed."**

It is now two for two and still unbuilt.

And say what that costs, because writing the recommendation is the part everybody does. The first bug was found by accident. The second was found by accident. **The project therefore has no estimate of how many it has not found,** because it has never once looked on purpose. Every number in the next two seasons rests on files that were checked only in the sense that nothing about them shouted.

What would count against this

1. **A corpus check that would have caught both.** Design it and you have improved on the project. Word counts against a published edition would catch the first. Asserting an expected row count would catch the second.
2. **These being the only two.** Nobody knows. **No systematic check has ever been run,** so the honest answer to "how many more are in there" is that the question has not been asked.

Handing off

The file is the right file. **Episode 6 asks what a word is,** which sounds trivial and produced a threefold error on the highest-scoring text in the corpus.

Questions

1. **Design the integrity check** that catches both bugs, in one paragraph.
2. Both produced plausible output rather than an error. **Coincidence, or a property of this kind of code?**
3. **How many corpus bugs are still in there?** Justify your number.
4. The marker was in the first line of output and was read past twice. **What does that tell you about reviewing your own diagnostics?**

Collecting episode 5

The file is now the whole file. **This episode is about turning that file into things you can count**, which sounds like the boring part and is where a threefold error lived.

Start from what a computer actually receives. **It does not get words**. It gets one long run of characters, and somebody has to write down a rule for where each word begins and ends. That rule is called tokenisation, and every count in every text-analysis result you have ever read was produced downstream of one.

The clinical anchor

Units.

The most dangerous number in medicine is a **correct number in the wrong units**. Nobody argues about the arithmetic. The arithmetic was fine. The measurement was fine. The instrument was calibrated and the technician was competent, and the number that came out is simply not the quantity anybody thought it was.

Notice what makes that class of error survive. **Nothing about it looks wrong**. A value in the wrong units still arrives in a plausible range, still varies sensibly, still responds to the things you would expect it to respond to. It fails no check that operates on the number.

The character anchor

Spinal Tap's Stonehenge.

The band asks for a Stonehenge monolith. Somebody sketches it on a napkin and writes 18 inches where they meant 18 feet. **Every subsequent step is performed correctly**. The drawing is measured accurately, the fabricator builds exactly what was specified, the crew flies it in on cue.

An eighteen-inch monument descends onto the stage, and dwarves dance around it.

Nobody in that chain did their job badly. The units were wrong at the first step and every competent operation afterwards carried the error faithfully to the audience. That is tokenisation. It happens once, early, before anything anybody would call analysis, and everything careful you do later is careful about the wrong quantity.

The device: one sentence, three tokenisers, three answers

Put this on screen. Take a line of Sophocles and count the word `king`.

```
raw text:      "O king, the king is dead," said the king's herald.

split on spaces: king, king king's          -> 1 match
strip punctuation: king king king s         -> 3 matches
strip + decode: king king king s            -> 3 matches, clean
tokens
```

Walk the first row. Splitting on spaces is the obvious rule and the one everybody writes first. It produces the token `king,` with the comma attached, and `king,` is not the same string as `king`, so it does not match. **Two of the three kings in that sentence are**

invisible to the count, and nothing in the output says so. You get the number 1, and 1 is a perfectly ordinary number.

Same sentence. Same word. Three different answers, and none of the three is obviously wrong until you look.

Nobody publishes this step. Every paper says "we tokenised the corpus." **The tokenisation is where the result was decided, and there is no section of any paper where you can check it.**

Three defects, each of which changed results

HTML entities. The downloaded pages contained **3,037 undecoded entities**, the codes a web page uses for apostrophes and dashes. Each one became a junk token: not a word, not punctuation, but a short run of letters that the tokeniser counted as a word because it looks like one.

And here is why that is worse than it sounds. **Rates are hits divided by word count.** The junk never matches any bucket, so it adds nothing to the numerator, and it **inflates the denominator of every single score in the corpus.** Not one text. All of them, in the same direction, quietly, downward.

Punctuation. Splitting on spaces leaves punctuation attached, so `king,` and `king.` are not `king`. On Sophocles' *Oedipus* this undercounted the `marks` bucket **threefold: 20 hits where the true figure was 62.**

A threefold error, on the highest-scoring bucket, of the highest-scoring text in the corpus, caused by a comma.

Take the three qualifiers in that sentence one at a time, because their combination is the point. *Oedipus* is the flagship text, the one the theory most expects to score. `marks` is the bucket it scores on. And the error was in the direction that hides the result rather than manufactures it. **The project's best case was being suppressed by punctuation**, and had nobody looked, the write-up would have reported a weaker finding and been believed.

The fix in both cases is one line, replacing a naive split with a rule that extracts letter-runs and lowercases them. Which is exactly why it deserves an episode: **the correction is trivial and the consequence was not.** There is no relationship between how hard a defect is to fix and how much it moves your results, and the intuition that the two go together is wrong in both directions.

The one thing done right

The Bible text used in later rounds was recorded with its source URL **and its sha256 checksum**, and the pre-registration says:

"If a future download has a different checksum, **that is a changed sample and needs saying out loud.**"

A checksum is a short string computed from the file's contents, such that **any change to the file at all produces a different string.** One character edited, one paragraph restored

by an editor, a different printing silently substituted by the source site: all of it shows up, without anybody having to remember what the file used to say.

One line, and it makes the sample checkable forever. Which sets up the sting: **nothing else in the corpus has one.** The single text with the strongest guarantee is the one that arrived last, and every text scored before it rests on the assumption that the internet does not change.

What would count against this

1. **The entity bug biasing all texts equally.** If it did, comparisons between texts survive. **Nobody has checked whether it was uniform,** and it would not be, because different sources had different entity rates.
2. **The tokeniser choice not mattering.** Easy to test: rerun with three tokenisers and compare. **Never done.**

Handing off

The text is right and the tokens are right. **Episode 7 is about the moment somebody nearly changed which texts were in the corpus, for entirely good reasons, after seeing the results.**

Questions

1. You have a corpus. **Write the five checks you run before computing anything.**
2. The entity bug inflated all denominators. **Does a uniform bias matter if you are comparing texts to each other?** Careful.
3. **Why does no methods section ever contain the tokeniser?**
4. Run the three-tokeniser test on any text you have. **How big is the spread?**

This is the strongest episode in the season and probably in the series. It is also the one to open the whole channel on, because it needs no theory at all.

Collecting episodes 5 and 6

The file is the whole file and the tokens are right. **Two bugs found, both of the quiet kind**, both producing plausible output rather than an error.

This episode is a third thing of the same shape, and it is the worst one, because **nothing about it is a bug at all**. Every step is correct.

The situation, told as a story

July. A prediction has already failed and the failure is published. The prediction was that mythography would score **lower** than tragedy, because the theory says myth conceals the mechanism and tragedy half-exposes it. **It lost by 0.04**, which is nothing at all, and a loss by nothing is still a loss and was reported as one.

Weeks later, while fixing the truncated-play bug from episode 5, the audit turns up something else. **A page of a text that the fetcher had missed entirely.** Real, legitimate, part of the corpus, simply never downloaded. Nobody left it out. A script did not follow a link.

And the missing page belongs to the genre that won.

The device: five true statements

Put these on screen one at a time and ask the audience to reject one.

1. The page really is part of that text.
2. The fetcher really did have a bug, and the bug is now fixed.
3. Fixing bugs in your data pipeline is **what you are supposed to do**.
4. The corrected corpus is **more accurate** than the one you scored.
5. Therefore rescore with the corrected corpus.

Every single statement is true. Every one is what a careful person does. Which one do you reject?

Now go along the row and try each rejection out loud, because the audience needs to feel the trap close rather than be told about it.

Reject 1 and you are claiming a page of Diodorus is not part of Diodorus. **Reject 2** and you are pretending a bug you found and fixed was not a bug. **Reject 3** and you have adopted the position that data pipelines should be left broken, which nobody holds. **Reject 4** and you are saying the corpus missing a page is the more accurate one, which is simply false.

Four of the five are unrejectable. **And here is what happens when you follow all five:**

MYTHOGRAPHY	13.95	->	13.67	vs	TRAGEDY	13.88
prediction:	FAIL	->	PASS			

A failed prediction becomes a passing one.

And the sentence that is the whole episode

From the correction file, written by the person it happened to:

"It would have been **entirely defensible as a bug fix**, and it is exactly the move pre-registration exists to make impossible."

And then:

"None of it felt like cheating."

Sit on that. Not "I was tempted and resisted." Not "I noticed what I was about to do." **The step did not register as a temptation at all**, because every link in the chain is a link you are supposed to follow, and the person following them was doing maintenance rather than analysis.

That is why the usual defences are worthless here. Integrity does not help, because nothing presented itself as a choice about integrity. Care does not help, because the whole sequence **is care**. **Being the kind of person who fixes bugs is precisely what walks you into it.**

That is what Season 2 episode 3 meant by "the shaping is invisible from the inside."
This is what it looks like when it happens.

The character anchor

Walter White, and specifically the early decisions rather than the late ones.

There is no scene in which he chooses to become a criminal. **Every individual step is a reasonable response to the situation the previous step created**, and each one is the sensible move for a man in that position at that moment. Ask somebody to point at the decision that did it and they cannot, because there isn't one. There is a chain, and every link is locally defensible.

The five statements are that chain, compressed into one afternoon. The difference is that in the show the audience can see where it goes, and in a research project nobody can, because the destination is just a number that came out slightly different.

What was actually done

The page was fetched. **It is on the disk. It is excluded from the analysis.**

And it stays excluded, permanently, from this round. It is available **only to a future round whose predictions are committed before it is scored.**

So the resolution is to reject statement 5, and only statement 5, and only for this round. **Notice how strange that is.** You knowingly compute your headline number on a corpus you have already established is missing a page, while holding the corrected corpus on the same disk. **The instrument is deliberately kept less accurate than it could be**, and it is the right call, and nothing about it feels right while doing it.

And a rule was written that had not existed before:

The sample is part of the registration, not just the instrument.

Everyone knows you cannot change your hypothesis after seeing the data. **Far fewer people know you cannot change your data either**, and this is why. The lexicon was frozen, the metric was frozen, the predictions were frozen, and the corpus was left as the one live variable in a system built entirely out of frozen ones.

The clinical anchor

Post-hoc exclusion criteria.

A trial where the eligibility rule is refined after the interim look. **Every individual refinement is medically reasonable**. Each one has a paper behind it, each was proposed by somebody competent for a stated clinical reason, and the refined population is genuinely the one the drug is for.

And the trial is dead anyway, and everybody in the field knows it is dead, and **nobody can point at the specific decision that killed it**. That is the signature. A trial killed by one bad decision can be discussed. This kind is killed by a sequence in which every element is defensible, so the argument about it never reaches a conclusion and the result simply stops being cited.

And now the uncomfortable follow-up, which belongs here and not in a later season

That rule was written in July. **Two weeks later the same project broke it**.

A minimum text length of 800 words had been declared, and had explicitly excluded three short texts by name. The round-4 sample file **contains no length rule at all**, and those three texts are back in.

TEXT	WORDS	ROUND 2	ROUND 4
hyginus_fabulae_2	613	excluded	in
hyginus_fabulae_3	689	excluded	in
hyginus_fabulae_4	777	excluded	in

Recomputed effect: the myth median moves 0.547 to 0.561, one counterexample count goes 7 to 6. **No verdict changes**.

And that is exactly the point. The discipline is not "did it matter." It is "**was it declared**." A rule that only binds when somebody is watching is not a rule, it is a habit, and habits lapse in fourteen days.

Put the two halves of the episode next to each other, because together they say something neither says alone. **In July the project refused a change that would have helped it, at real cost, on principle**. In August the same project silently dropped a rule it had written itself, in a way that happened to be harmless, and nobody noticed for weeks.

The same person, the same repo, two weeks apart. **So the July refusal was not a character trait**, and treating it as one is the mistake. It held because it arrived as a visible decision with a written rule attached. The August lapse happened because nothing arrived at all.

Which is the refrain, in its most concrete form. A rule that depends on your remembering it is a check that cannot fail, and a check that cannot fail prints reassurance. The length rule was in a document. It was not in the code.

What would count against this

1. **The Diodorus exclusion being superstition.** A real argument exists: the page belongs, excluding it makes the corpus knowingly wrong, and pre-registration should not force you to publish a number you know is inaccurate. **Make that argument at full strength before rejecting it.**
2. **The length-rule lapse being harmless because nothing moved.** Also arguable. **Argue it, then notice that you can only say it moved nothing because somebody checked afterwards,** which is not available in advance.

Handing off

The corpus is fixed, frozen, and its rules are written down.

Episodes 8 to 10 turn to the other half: the word-lists, and to the day somebody nearly repaired one to make a failing prediction pass.

Questions

1. **Which of the five statements do you reject?** You must reject one. Say which and why.
2. Is the Diodorus exclusion the right call, or superstition? **Argue both sides before answering.**
3. The length rule lapsed and changed nothing. **Better, worse, or neither?**
4. **Name a mechanism that would catch a lapsed rule.** Not "be more careful."

Collecting episode 7

The corpus is settled and its rules are written down. **Now the other half of the instrument: the word-lists**, and the first decision is one every text-analysis tutorial gets wrong.

The temptation

Every tutorial says the same thing. **Reduce words to their stems**, so **murder**, **murdered** and **murdering** all count as one concept.

And the reason it is recommended is a real reason. Without it you are counting strings rather than ideas, and a text that says **murdering** three times looks identical to a text that never mentions killing at all. Stemming is one line of library code, it costs nothing, and it fixes that.

So understand what a stemmer actually does before deciding against it. **It chops suffixes according to rules about letters**. It does not know what any word means. It knows that a word ending in **-ing** is probably a form of the word without the **-ing**, and it applies that everywhere, at speed, without hesitating.

This project refused it entirely. No prefix matching anywhere.

The device: four stems, and what each one drags in

Put the counts on screen. These are from the actual corpus.

YOU STEM	YOU ALSO CATCH	HOW MANY TIMES
tear-	"tears", which is weeping, not rending	441
rend-	"render", "rendered", "rendering"	28
torn-	"tornado"	
stone-	"stone" the material, against "stoned" the punishment	211

Go down the rows, because each one fails differently.

rend- catching "render" is the innocent case. 28 hits, spread across the corpus, meaning nothing in particular. That is noise, and noise you can live with.

torn- catching "tornado" is the funny case, and it is the one that gets used in talks, and it is the least dangerous of the four. Nobody will ever be misled by it, because the moment you look at a hit you can see it is absurd.

stone- at 211 is where it stops being funny. The bucket wants stoning, the execution. The corpus is full of stone the material, stone walls, stone steps, stone thrown in anger with nobody dying. **The false hits are in the same semantic neighbourhood as the true ones**, which means you cannot spot them by scanning a list. You have to read every occurrence in context, which is the labour the stemmer was supposed to save.

And then read the first row slowly. The bucket exists to catch a body being torn apart by a crowd. **Stemming hands it four hundred and forty-one instances of people crying**.

`tear` and `tears` are the same letters and opposite events. No rule about suffixes can separate them, because there is nothing in the letters to separate.

Why the first row is not just the biggest number

Those 441 hits do not scatter randomly. **They concentrate in exactly the texts you are most interested in**, because tragedies are full of weeping.

The error correlates with the hypothesis, and that is the worst property an error can have. Spell out why, because most people rank errors by size and size is the wrong axis.

A random error adds noise. Noise makes a true effect harder to find, so it works against you, and a result that survives it is stronger than it looks. **A correlated error adds signal shaped like your hypothesis.** It works for you, it grows as your effect grows, and it survives every check that asks how large the effect is, because the effect really is large. It is simply not yours.

A big random error is a problem you can see. A correlated error of any size is a result you cannot tell from a discovery.

The clinical anchor

Specificity.

A screening test that flags everybody has perfect sensitivity and catches every case in the population, and it is useless, because a positive result carries no information. Nothing was learned that you did not know before running it.

Stemming is a screening rule with excellent sensitivity and terrible specificity, applied to a question where the false positives cluster in the signal. It is the version of the useless screen where the extra positives are not even distributed evenly.

The character anchor

Drax.

He hears every word correctly and takes it entirely on the surface. Nothing goes over his head; his reflexes are too fast, he would catch it. The failure is not inattention and it is not stupidity, because **his processing of the form is perfect**. He has no model of what the form is for.

A stemmer is Drax. Letters go in, the suffix rule applies, and the meaning was never consulted at any point. It will do this a hundred thousand times without tiring, without doubting, and without once producing the hesitation that would tell you it had gone wrong.

What was done instead

Every concept lists its surface forms explicitly. So `tear_apart` is exactly `tore`, `torn`, `tear`, `tearing`, and nothing else. Written out by hand.

The cost is that adding a concept means enumerating it, and the project's own code comment says:

"Explicit lists mean **every match is auditable**. The cost is that adding a concept means writing out its forms; **that cost is the point**."

Read the last clause again, because it is the sentence the episode exists for. The effort is not a regrettable side effect that a better tool would remove. **The effort is what forces every form to pass through a decision**, and a decision leaves a record that somebody else can disagree with.

The trade, stated honestly, because explicit lists are not simply better

	EXPLICIT LISTS	STEMMING
false positives	few , and each one traceable	many, and invisible
false negatives	many . Any form you forgot is silently missed	few
auditable	yes . Open the file and read it	no
effort	high	one line

Look at the second row before congratulating anybody. **Explicit lists are worse on false negatives, and not slightly**. Every form nobody thought of is missed in silence, and the misses do not announce themselves the way a "tornado" hit does. The instrument is under-counting all the time and nothing on the screen says so.

So the honest comparison is not accuracy against accuracy.

Explicit lists trade a loud, checkable error for a quiet, uncheckable one. They are not more accurate. **They are more inspectable**, and in a project whose entire defence is "go and check," that is the right trade **and it should be presented as a trade rather than as a virtue**.

And notice which way that cuts against the project's own interests. The choice that makes the instrument inspectable is the choice that makes it **miss more**, so the discipline here is paid for in results, not in praise.

What would count against this

1. **A form the lists miss that carries real signal**. Certain. **The question is whether the misses correlate with the hypothesis**, and nobody has checked.
2. **Stemming giving the same answer**. Testable in an afternoon: rerun with a stemmer and compare. **Never done**.
3. **A modern language model solving it**. It would fix the coverage problem and **introduce an unauditable one**, which is a different trade, not a free win.

Handing off

The lists are explicit. **Episode 9 is the day somebody went through them deleting things**, and the day a reasonable fix was refused.

Questions

1. **Which error would you rather have**, the loud checkable one or the quiet uncheckable one? Does it depend on the claim?

2. `tears` and `tearing` are one string family and opposite meanings. **How many other pairs like that are in English?**
3. The 441 false hits concentrate in tragedies. **Why is a correlated error worse than a random one?**
4. **Would a language model solve this?** What new problem does it introduce?

Collecting episode 8

The lists are explicit and hand-written, which is the choice episode 8 defended. Now collect the cost of that choice. **A hand-written list means one person chose every single word in it**, and that person is not neutral. He built the instrument because he expects it to find something.

Every word he adds is a word he thinks belongs. Every word he leaves out is a word he judged did not. There is no mechanism anywhere in the procedure that pushes back on him.

This episode is him going through it deleting.

What came out, with counts

REMOVED	HITS IT WAS CONTRIBUTING	WHY
bare stone, stones	211	nearly all the material. A stone wall, a stone tomb
bare cast, drive, driven	749 combined	"cast a glance", "driven by the wind"
tears, tearful	462	weeping
render, rendered	28	not rending
ravishing		almost always "ravishing beauty" in this corpus

Read the second row, because it is the largest and the least dramatic. **749 hits from three ordinary English verbs**. Not exotic words, not mistakes anybody would laugh at. **cast** and **drive** are simply common, and common words accumulate enormous counts in a corpus of this size while meaning nothing relevant on almost every occasion.

Roughly 960 hits deleted from an instrument whose author wanted the numbers to go up. Every single removal lowers every single score, in every text, including the ones he most wanted to score high.

That is the episode's evidence, and it is a kind that cannot be faked. You can claim you were careful. **You cannot claim you deleted 960 hits from your own case unless the file shows it.**

And notice why the direction is what carries it. An author tuning an instrument towards his hypothesis **adds** vocabulary that fires in the texts he wants to win. That is the natural motion, it costs nothing, and it leaves a file that looks exactly like a thorough one. **These 960 are the opposite motion**, and the opposite motion is expensive in the only currency the project cares about.

The craftsmanship, which is the interesting part

cast and **drive** did not simply go. **Girard's stereotype is expulsion, and English builds expulsion with particles.**

So they came back as phrases: **cast out**, **cast forth**, **cast down**, **drive out**, **drove out**, **driven from**, **thrust out**, **hunted out**.

The bare verb is noise. The verb plus its particle is the concept. That is a claim about **English**, not about Girard, and claims like it decide results far more often than anybody admits.

Sit with that for a second, because it is a whole category of hidden decision. The pre-registration froze a theory of persecution. **It did not freeze a theory of English**, and one was needed anyway, and one got supplied on the spot by whoever happened to be editing the file. Nobody pre-registers their linguistics. The grammar you assume is invisible in a way the theory you announce is not.

The device: the refusal, and why it is not a statistical decision

Here is the situation. A prediction has failed. Specifically, **Girard's single most important example scored zero** on the crimes bucket, because Ovid describes the killing as **tearing and rending** while the bucket contained **murder**, **parricide**, **pollution**.

The obvious fix, and it is written in the project's own next-steps list: add the tearing vocabulary to the crimes bucket.

It would work. It would fix the failure. It is a coverage gap, coverage gaps are meant to be fixed, and this one is real: the bucket genuinely does not contain the words Ovid genuinely uses. **Nothing about that repair is dishonest on its face**, which is exactly the problem, because nothing about the repairs that ruin projects ever is.

It was refused. And the reason is not statistical.

"In Girard the crime stereotype is **what the victim is accused of**; the tearing is **what the mob does**. Tearing belongs in violence. **Putting it in crimes would have corrupted the bucket to pass a test.**"

The vocabulary went into the **violence** bucket, where it belongs, **and where it does not help the failed prediction at all.**

So look at what actually distinguished the two options. Both are the same edit: take the tearing words, put them in a bucket. **The words are identical, the labour is identical, and the only difference is the destination.** One repair fixes a coverage gap. The other manufactures a pass. There is no test on the words themselves that separates them, and there is no number anywhere in the project that could have decided it.

The instrument was one line of reasoning away from being repaired to make its own failing example pass, and what stopped it was a definitional distinction from Season 1 episode 7, not a number.

That is why Season 1 says that distinction twice, in episode 7 and again in episode 9. A rule written in advance can refuse you. The same rule, articulated on the day it is needed, refuses nothing, because a person under pressure to pass will articulate whichever version of the rule lets him pass and will believe it while doing so.

The clinical anchor

Moving a patient's diagnosis to make the pathway work.

Everybody knows what that is. Everybody knows how reasonable it feels at the time, and that the reasoning offered for it is usually correct as far as it goes: the patient really does have that feature, the pathway really is the one that will help. **The reclassification is defensible in every individual case and the audit still has to catch it**, and the audit only can if somebody wrote down what the categories meant **before** the patient arrived.

The character anchor

The two fake rules in Death Note.

Light is the prime suspect, and the evidence against him is going to hold. So he writes two extra rules into the notebook itself, in the same hand as the real ones, on the same pages, saying things that clear him. The investigation then reads the rulebook, reasons impeccably from it, and reasons its way to the wrong man.

The trick works for one reason only. The notebook carries no record of when any rule was written. There is nothing in the object to distinguish a rule that was always there from a rule added by the person the rule exonerates.

A lexicon is a rulebook. The tearing repair would have been a rule added to clear the suspect, in the same hand, on the same page, and every analysis downstream would have reasoned correctly from it.

What would count against this

1. **The tearing vocabulary genuinely belonging in crimes.** Argue it: the mob believes it is punishing, so its action is an accusation enacted. **It is not a stupid argument.** The counter is that Girard's own text separates them.
2. **The 960 deletions being over-correction.** If `stone` sometimes really is the punishment, deleting it costs real signal. **Nobody measured how much.**
3. **Deletion being a form of tuning too, and this is the sharpest version of the objection.** Removing words that fire mostly in your controls raises the contrast exactly as effectively as adding words that fire in your targets, and it looks like restraint while doing it. **Whether these 960 hits fell evenly across the tiers or landed disproportionately on the control texts was never measured**, and the direction-of-sacrifice argument above quietly assumes they fell evenly.

Handing off

The lexicon was revised **after** seeing a failure. **Episode 10 is about the three characters that make that survivable.**

Questions

1. The tearing vocabulary would have fixed the failure. The category rule said no. **Which authority should win?**
2. **Deleting 960 hits lowers your own scores. What makes someone do that?**

3. "Bare verb is noise, verb plus particle is the concept." **Fact about English or convenient rule?** How would you check?
4. Argue the case **for** putting tearing in crimes, at full strength.
5. The right fix and the cheating fix were the same words in different buckets. **Name a decision in your own field with that shape**, where nothing in the action distinguishes the two and only a prior rule does.

Collecting episode 9

Episode 9 revised the lexicon, and it revised it **after seeing a failure**.

State the problem precisely, because the loose version of it is wrong. The loose version says that revising an instrument after seeing results is cheating. That would forbid every bug fix ever made, and no instrument in any field would survive contact with its first real data. Episode 9's changes were made for good reasons and this episode does not take them back.

The precise version is that **two different kinds of change were made in the same edit**, and afterwards nobody could tell them apart.

Did the numbers improve because **bugs were fixed**, or because **new words were added that happened to help**?

Those are not two degrees of the same thing. **A bug fix says the instrument was failing to do what it always claimed**. The theory is untouched. **Adding a concept says the instrument now claims something different**. What is being tested has changed, and any comparison to a result from before the edit is a comparison between two instruments.

And a reader looking at the revised lexicon sees one file, one date, one list of words, with nothing on the page indicating which line came from which motive.

That is not a flaw in the results. It is a hole in the record, and holes in the record cannot be repaired by argument afterwards, because there is nothing left to argue from.

The device: three characters that make the edit reversible

Every entry in the repaired lexicon carries a tag.

TAG	MEANS	IS IT A THEORY CHANGE?
v1	in the original pre-corpus lexicon, unchanged	no
inf	an inflection of a v1 word. A matching bug fix	no. Adds no concept
v2	a genuinely new concept added in the revision	yes

Take them one at a time, because the middle row is the one doing the work.

v1 is the frozen state. Whatever was written in episode 2, before any text had been scored, before anybody knew what would happen.

inf is an inflection of something already in v1. `murder` was in the original list and the matcher was failing to catch `murdered` and `murdering`. Adding those forms adds no concept whatsoever. **The instrument always intended to catch that word and was simply failing to**, which is the definition of a bug rather than a revision.

v2 is a concept that was not there. Something the pre-corpus lexicon did not contain and would never have caught however perfect the matching had been.

And then the part that turns a label into a check. **The scorer takes a flag, --no-new, which drops everything tagged v2**.

Why a tag is not a changelog

Most projects handle this with a note. We revised the lexicon on such a date, here is what we changed and why. That note is prose, and a reader has to trust it, then reconstruct the old instrument in their head, then take on faith that the reconstruction matches what the code actually did.

The tag is executable. Any result in the project can be recomputed as if the new concepts had never been added, by anybody, without asking permission and without the author's cooperation.

A changelog is a **claim** of separability. A flag **is** separability. The point is not that the author was honest about the revision. **The point is that his honesty stopped being load-bearing.**

And put it against the refrain. A promise not to have tuned the instrument is a check that cannot fail, so it prints reassurance and nothing else. **--no-new can fail.** Run it, and if the results collapse without the new concepts, the project has been told something it did not want to hear, in public, by its own code.

The clinical anchor

A protocol amendment log.

Amendments are not forbidden. Trials get amended constantly and for excellent reasons. **The requirement is that each one is dated and separable**, so a reader can reconstruct what the trial looked like before it.

Which is why the failure mode here is worse than it sounds. **A trial with undocumented amendments is not a trial with a flaw. It is a trial you cannot read.** A flaw you can weigh. An unreadable record leaves you with a number and no way to find out what produced it.

The character anchor

Leonard in Memento. He cannot form new memories, so he keeps his findings on polaroids and tattoos, which is a perfectly sensible instrument for a man in his situation.

And it has no provenance tags. Nothing on any note records who he was when he wrote it, or what he had just been told, or which notes were written before which. So he cannot distinguish a fact he established from a fact somebody fed him, and the record he built to protect himself becomes the mechanism by which he is steered. **He is not lying and he is not careless.** He is unauditable, including to himself, and that turns out to be enough.

And here is what the tags revealed, which is the reason they earn an episode

The v1 headline finding had been: **"each bucket is dominated by one generic word."** It was the project's first real observation about its own instrument, and it mattered, because a bucket that is mostly one word is not measuring a concept. It is measuring a word, and reporting the result under the concept's name.

Then the matching bug was fixed, and the tags let somebody ask what that finding looked like without it.

BUCKET	TOP WORD'S SHARE, V1	AFTER THE FIX
crisis	68%	60%
marks	54%	44%
violence	36%	21%
crimes	43%	12%

Two of the four buckets stopped being dominated by one word once the matching was fixed. Crimes went from 43% to 12%.

The mechanism is worth spelling out, because it explains the direction of every row. The old matcher caught `murder` **138 times out of 297 real occurrences**. The forms it missed were the inflected ones. So the base form was the only form being counted reliably, and of course it dominated the bucket. **The dominance was a fact about the matcher, not a fact about the corpus.**

So half of the project's own first major finding was a bug in its own code, and the tags are the only reason anyone could tell.

Then look at which half survived. **Crisis is still 60%** `death` and `marks` **is still 44% king**. So "dominated by one generic word" is not a property of the method and never was. It is a per-bucket fact, true of two buckets and false of two, which is a worse outcome than a clean survival or a clean death because it gives you no rule to carry forward.

And the surviving 44% is the load-bearing one. It is the number Season 1 episode 8 is built on, and the one Season 4 episode 17 tests by deleting the word and re-running everything.

Without the tags, none of that is detectable. There would be no correction to announce. You would simply have a better lexicon, and a claim that stopped being repeated, and nobody would ever have said a false thing out loud.

What would count against this

1. **inf not really being "no new concept."** Argue it: inflections smuggle in meaning. `murdering` is not `murder`, it is ongoing, and a bucket that gains the progressive forms may be catching a different aspect of the event. **A real objection and the project does not address it.**
2. **The tags being applied inconsistently.** Checkable by reading the file, which is the point of putting them there.
3. **The tags being assigned by the same person making the edit.** Nobody independent decided whether a given form was an inflection or a smuggled concept, so the separation is only as good as one author's judgement at one sitting. **The check is executable and the classification behind it is not.**

Handing off

The instrument is built. **Episode 11 is the first number it produces**, and the one property of that number causes every major failure in the next two seasons.

Questions

1. **Without tags, how would you ever separate the two effects?**
2. Half the v1 finding was an artifact of the code. **What in your own work is a property of your tools rather than your subject?**
3. **Is `inf` really no new concept?** Argue that inflections smuggle meaning.
4. Two buckets kept their dominant word and two lost it. **What, if anything, are you entitled to conclude from a finding that holds in half your cases?**

Collecting episode 10

The corpus is right, the tokens are right, the lexicon is written, debugged, de-noised and tagged.

Now you count. And the counting has one property that costs the project two entire seasons.

The metric

$$r(\text{text}) = 1000 \times \text{hits} / \text{total words}$$

Hits per thousand words. That is the whole thing.

And the denominator is not decoration. A raw count is worthless across texts of different lengths, because a long text contains more of everything, including whatever you are looking for. Dividing by length is the minimum required to make two texts comparable at all, and every field that counts things arrived at the same fix independently.

The clinical anchor, exact

Prevalence. Cases divided by population.

You already know why the denominator matters and why raw counts mislead: **a bigger hospital has more infections and is not dirtier.** Same idea, no translation needed.

And the character version: a batting average. Everybody accepts instantly that you divide by opportunity, and nobody has ever needed the concept explained.

So the rate is the right instinct, correctly applied, and this episode is not about a mistake. **It is about what the correct instinct costs.**

The device: shuffle the book

Do this on camera and make it physical.

Take the *Bacchae*. Seventeen thousand words. Cut it into individual words. Put them in a bag. Shake the bag. Pour them out in whatever order they come.

Now score it.

The number is identical. To the last decimal place.

Not close. Not approximately. **Identical**, and it is worth being clear that this is not a limitation of the implementation, because it follows from the arithmetic with no room for anything else. The numerator counts hits, and shuffling does not create or destroy a word. The denominator counts words, and shuffling does not create or destroy those either. **Both terms are functions of which words are present and of nothing else.**

The rate cannot tell the difference between Euripides and a bag of the same words.

Why that single fact matters more than anything else in this season

Write it on the board and leave it there for two seasons:

A rate treats a text as a bag of words. So it can never test any claim about **where** something happens, **when** it happens, or **what it happens near**.

Take a moment on what is inside "order," because it is easy to hear as a small thing. Order is adjacency, sequence, scene, and episode. It is what happens next, and what happens to whom, and what one passage does to the passage after it. **Order is the whole of narrative**, and a rate discards all of it in a single division.

And now recall what the theory actually claims. **Season 1 episode 10**: the four stereotypes converge **on one passage**, because one event produced all four.

Girard's claim is about a moment. A rate cannot represent a moment. It cannot represent anything happening anywhere.

That mismatch is the cause of every major failure in Seasons 3 and 4. Not a bug, not a bad word-list, not too small a corpus. **The instrument and the claim are about different kinds of thing**, and it takes eight rounds and four instruments to fully absorb it.

Say the consequence out loud, because it is the part people resist. **No amount of additional data repairs this.** A hundred more texts, a cleaner corpus, a better lexicon, a more sophisticated test: every one of those improves the precision of a quantity that does not contain the claim. **You can measure a bag of words as accurately as you like and still never have measured a moment.**

The second thing it discards, which is subtler

A rate rewards density of event. A text where every word is plot carries more hits per word than a text with description and lyric in it, and the difference has nothing to do with persecution. It is a fact about how the text was written.

Measured later: mythography runs **28.65 neutral event-words per thousand** against 21 to 23 for everything else. **The densest narrative prose in the corpus**, and the reason is structural rather than accidental. Apollodorus is a handbook. He records what happened and drops everything a poet would put around it, so a far higher proportion of his words are doing narrative work.

The problem is that the denominator counts all words equally while the numerator only ever fires on event vocabulary. **Strip the lyric out of a text and its rate rises without a single new event being described.**

And in Season 3 episode 7 mythography **ties tragedy**, which nobody predicted, and which Season 1 episode 17 offered two readings of.

And what a rate is genuinely good for, because this episode is not an attack

Say this before the objections, or the audience will conclude the metric is worthless and that is not the finding.

A rate is the correct instrument for any claim about how much. How much persecution vocabulary does this genre carry. Does mythography carry more than tragedy. Is this text unusual for its kind. **Every one of those is answerable and worth answering**, and the rate answers them with a precision no reading of the texts could match.

It also has a property nothing else in this project has: **it is completely transparent**. Anybody can recompute it with a word-list and a calculator, there are no free parameters to tune, and two people scoring the same text will get the same number to the last decimal.

The problem is not that the rate is a bad measurement. It is that it is a good measurement of the wrong quantity, and those two failures look identical from inside the spreadsheet.

Which is why the fix in Season 3 is not a better rate. It is a different question, asked with a window and a peak, and it takes two failed rounds to get there.

What would count against this

1. **Position not mattering.** If the stereotypes really are diffuse properties of whole texts rather than features of episodes, the rate is the correct instrument and Seasons 3 and 4 are chasing something that was never there. **That is a live possibility and nobody in the project entertains it**, which is a bad sign in itself. The reading that would make eight rounds of work unnecessary is the one nobody argued for.
2. **A better denominator existing.** Divide by event-words instead of total words? Season 3 episode 8 computes exactly that and **it flips a result**, which is why it cannot be claimed.
3. **This episode overstating the loss.** A rate is blind to order, and it does not follow that order carries the signal. **A text drenched in all four stereotypes at every point would be found by a rate and found correctly.** The argument here is that the rate cannot test the conjunction claim, not that it tests nothing.

Handing off

You have four numbers per text. **Episode 12 asks how to combine them**, and the two options encode two different theories.

Questions

1. **Name three things a rate per thousand throws away.** You have two.
2. Shuffling the book does not change the number. **What class of claim can that number never test?** State it precisely.
3. **What would you divide by instead of total words?** What breaks?
4. Is it possible the stereotypes really are diffuse? **What would show that?**

Collecting episode 11

Episode 11 produced four numbers per text, one per stereotype, each blind to order. **This episode is about turning four numbers into one**, which you have to do, because you cannot rank texts or test a prediction against a quadruple.

And how you combine them is not a technical choice. The two options encode two different theories, and picking one is picking what you think Girard is claiming.

Which is why the episode is worth ten minutes on something that looks like arithmetic housekeeping. **The theory gets decided here, in a line of code, by somebody who may not notice he is deciding it.**

The two options

$$SUM = r_{crisis} + r_{crimes} + r_{marks} + r_{violence}$$

$$MIN = \text{the smallest of the four}$$

SUM is accumulation. A text scores high by having a lot of any of it, and one enormous bucket can carry the other three on its back. Say that property by its name, because the next episode turns on it: **SUM is compensatory.** Strength in one place pays for absence in another, and the total does not record which happened.

MIN is conjunction. A text scores only as high as its **weakest** stereotype, and nothing a strong bucket does can raise it. Drench a text in violence with no accusation anywhere and it scores **zero**.

The device: same four numbers, two verdicts

Put two invented texts side by side.

	CRISIS	CRIMES	MARKS	VIOLENCE	SUM	MIN
Text A	2.0	2.0	2.0	2.0	8.0	2.0
Text B	0.0	0.0	0.0	8.0	8.0	0.0

Identical on SUM. As different as two texts can be on MIN.

Text A has all four stereotypes at once. **Text B is a battle scene with no victim, no accusation and no crisis in it at all.** SUM says they are the same text. **MIN says one of them is not a persecution text.**

Notice that the table contains no new information beyond the four columns. Both metrics see identical inputs. **Everything separating the two verdicts happens in the choice of operation**, which is why calling it a technical detail is the mistake the episode exists to prevent.

And be clear that SUM is not stupid. It answers a real question, which is how much persecution vocabulary a text carries. **That question is simply not the one being asked**, and the two are easy to confuse because a text that satisfies the second usually satisfies the first as well.

The clinical anchor, exact

Diagnostic criteria requiring N of M against criteria requiring all of them.

SIRS needs two of four and catches everything that walks through the door. A definition requiring all four is specific and misses cases. **Same trade, same mathematics, and every clinician has an opinion about it.**

Notice that neither is wrong. The N-of-M version exists to make sure nobody is missed, and it earns its keep in a triage setting where the cost of missing is enormous. **All-of exists to make sure the label means something**, and it earns its keep when somebody is going to build on the label later. The question is never which is better. It is what the label is for.

The character anchor, and you already have it

The relay team from Season 1 episode 10. Nobody thinks the fastest individual leg tells you about the team, because **the baton has to go all the way round**, and the claim "this team is good" is a claim about the **weakest** leg.

Episode 8 used it to explain what Girard means by *together*. **Here it stops being an illustration and becomes the arithmetic**, which is the whole move of this episode: MIN is that sentence written as an operation.

And it is the dropped baton that carries it, because a dropped baton does not lengthen the team's time. **It ends the race**, no matter how fast the other three ran. That is what taking a minimum does to a zero, and it is why the algebra and the sentence are the same object.

Why MIN is the right choice for this theory

Season 1 episode 10 said it: **the whole claim is conjunction**. Four faces of one event.

Follow what that requires. If the four stereotypes are four traces of one mechanism, then a text with three of them and not the fourth is evidence **against** the mechanism, not weak evidence for it. The theory does not predict lots of persecution vocabulary. **It predicts co-occurrence**, and a metric that scores highly on three is not measuring co-occurrence at all.

MIN is the algebraic form of the sentence "they appear together." SUM is the algebraic form of "there is a lot of this stuff around." **Only one of those is Girard's claim.**

So MIN was chosen and written into the first pre-registration as the primary metric.

And here is the correction, which was found by opening the book

That last sentence is too strong, and the series said it for a long time before anybody checked.

Girard's own decision rule, in *The Scapegoat* chapter 3, is this:

"Not all the stereotypes must be present: **three are enough and often even two.**"

That is N-of-M, and it is not MIN. MIN demands four of four and returns zero on three. Girard demands two.

So the honest account of this choice has three parts and the middle one is new.

the reasoning above	still good. Conjunction, not accumulation, is what makes it a theory
the attribution	wrong. Girard does not require all four, and nobody had read the sentence where he says so
what MIN actually is	a deliberate strengthening of the source's rule, which is a legitimate thing to do and a different thing to claim

Choosing a harder test than your source requires is defensible and often admirable. Presenting it as what the source requires is not, and the difference matters here because **the strengthening is what breaks in Season 3.**

Say it that way on camera. MIN is the right instinct and an overreach, at the same time, and Season 1 episode 10 has the full passage and the reason the overreach is more interesting than the error.

The reasoning should still be presented as good, without hedging in the voice, because the audience needs to feel a decision being made well before they watch it fail. **Just do not tell them Girard asked for it.**

And the trap, planted not sprung

MIN has a property nobody considered, and it is **arithmetic rather than literary.** Nothing about Girard is involved in it, and nothing about the corpus, and no amount of thinking harder about the theory would have surfaced it.

The number that does the damage was already planted in Season 1 episode 7: **crimes fires at 0.95 per thousand words, five times rarer than violence.**

Put those two facts next to each other on the board and leave them there. A metric that takes the smallest of four. A bucket that is five times rarer than its neighbours. **The audience now has everything needed to see it,** which is the point of planting rather than explaining.

Do not resolve this here. Let the audience feel the choice was right, because it was right. **Season 3 episode 2 takes it apart in one line of arithmetic that has nothing to do with Girard.**

What would count against this

1. **MIN being too strict.** If the four stereotypes are genuinely produced by one event, they should still vary in strength, and demanding all four be present at once may reject real persecution texts. **That is the case against MIN and Season 3 episode 1 is what it looks like when it happens.**
2. **A third option existing.** Geometric mean? Second-lowest? **Try to invent one before you are shown one.** MIN is one end of a family, and the project never looked at the middle of it.
3. **The conjunction being at the wrong scale.** MIN asks whether all four occur in the same **text.** Girard's claim is that they occur in the same **event.** Those are not the same requirement and episode 11 already said why a rate cannot express the second one.

MIN is the best available answer to a question the instrument cannot ask properly.

Handing off

MIN is chosen. **Episode 13 is the case that proves it was the right call**, and it is a dinner party.

Questions

1. **State in one sentence what MIN detects that SUM cannot.**
2. MIN is the minimum of four. **What happens when one of the four is rare?** Answer before Season 3.
3. **Invent a third way to combine four numbers** that encodes conjunction without MIN's weakness.
4. SUM answers a real question badly matched to the claim. **Name a summary statistic in your own field that answers the neighbouring question** and gets reported as though it answered yours.

Collecting episode 12

Episode 12 chose MIN over SUM on theory alone. Nothing in the data recommended it. The argument was that Girard's claim is about four things occurring together, so the metric has to be one that four together can satisfy and three cannot.

That is the weakest kind of justification available at the moment you make it. You are asserting that a piece of arithmetic matches a piece of theory, and there is no way to check the match from inside the argument. Every bad instrument in the world was also chosen for reasons its author found convincing.

This episode is the check, and it comes from a text that was in the corpus to be boring. It is the only kind of evidence that can vindicate a choice made on theory: a case the choice handles correctly that nobody had in mind when they made it.

Season 1 episode 18 read this passage as literature. This episode reads it as a test of the metric.

The finding

BOOK	SCENE	CRISIS	CRIMES	MARKS	VIOLENCE	SUM	MIN
7	Odysseus welcomed at the palace	0.30	0.00	6.21	0.00	6.51	0.00
18	Irus the beggar	0.00	0.00	5.49	2.15	7.64	0.00
22	the slaughter of the suitors	1.09	0.22	0.22	4.14	5.67	0.22

Read it one row at a time, because the ordering is the finding.

Book 7 is a dinner party. Odysseus arrives at the palace of Alcinous, is bathed, given clean clothes, fed, and questioned politely by his host. Nobody is accused, expelled or harmed. **Zero on crimes, zero on violence**, and both zeros are correct, because there is nothing in the scene for those buckets to find.

Book 22 is the massacre. Odysseus kills every suitor in the hall, and it is one of the most violent scenes in Greek literature.

On SUM the dinner party outscores the massacre, 6.51 against 5.67. And it does not only beat book 22. On the victim-marks bucket it outscores **every tragedy in the corpus**, so it beats *Oedipus*, it beats the *Bacchae*, it beats every play the instrument exists to catch. The slaughter comes seventh.

Why a hospitality scene scores as victim-marking

Nothing has gone wrong with the code here, and that is the part worth slowing down for.

Collect Season 1 episode 8. `marks` is **44% the word king**, and that was not a lazy choice. The episode ran both tests on it and the king passed both. He is outside the group, because a king has no peers, and he can be killed without anybody being obliged to avenge him, because there is no other king to do it. **In Girard the king is a victim whose execution has been deferred**, flagged there as tier K and still tier K here. So `king` belongs in `marks` by the theory the instrument is implementing.

Now put that word-list in front of book 7. Everyone in the room is a king, is addressing a king, or is describing one. The word saturates the passage for reasons of **setting**.

The false positive is not the lexicon misfiring. It is the lexicon working exactly as written, on a text where one stereotype is present at maximum density and the other three are absent.

That decides which repairs are available. A misfire you fix by editing words. This you cannot, because deleting **king** would delete a real victim mark and break the instrument in the other direction. **The problem is not in the word-list, so it cannot be solved in the word-list.**

What SUM does with a text like that

SUM adds the four buckets. 0.30 plus 0.00 plus 6.21 plus 0.00 is 6.51.

Look at what that arithmetic permits. **A single bucket can carry the entire score.** Half of book 7's buckets are empty and SUM does not care, because 6.21 in one column pays for both zeros with room to spare.

Episode 12 called that property compensatory, and book 7 is what it looks like outside an invented example. Strength anywhere buys weakness anywhere else, so a text with all four stereotypes at a moderate level and a text with one stereotype at an extreme level produce the same total. Once they have, **no later step can pull them apart**. The number threw the difference away at the moment it was computed, and everything downstream inherits a quantity that no longer contains the thing being claimed.

Book 22 is the demonstration. All four of its buckets are non-zero. **On the claim being tested it is the better text**, because the claim is that the four occur together and in book 22 they do. SUM ranks it below the dinner party anyway.

The payoff

Look at the MIN column.

MIN takes the smallest of the four. For book 7 the smallest of 0.30, 0.00, 6.21 and 0.00 is **0.00**. For book 18, **0.00**.

Notice how the 6.21 gets treated. It is not penalised, not down-weighted, not corrected. **It is never consulted**, because MIN stops at the weakest bucket. An empty bucket anywhere sets the score to zero no matter what the rest of the text is doing.

Book 22 scores 0.22. Small, above both, and the ordering is now the right way round.

The conjunction requirement kills the false positive automatically, with no patch, no exception, and nobody having to notice.

The load-bearing clause is the last one. Nobody wrote a rule about kings or about hospitality scenes, and nobody read book 7 in advance. The metric was chosen because of the shape of the claim, and the shape of the claim is what rejected the text.

Then consider the alternative. An instrument that has to be told about book 7 will also have to be told about the next book 7, **and no list exists anywhere of every text that will fool you**. A special case only ever covers the failure you have already met.

The clinical anchor, and it is episode 12's, now with a patient in front of it

Episode 12 put N-of-M criteria against all-of criteria and left it as a trade. **Book 7 is what the trade looks like when a real case arrives.**

Any-five-of-nine admits this patient. One florid feature at an extreme level, four soft ones, threshold met, diagnosis recorded, **and the chart does not preserve which route it took.** Everything downstream now treats that patient as equivalent to somebody who met the criteria the ordinary way.

All-of refuses the same patient on one absent feature, and refuses without anybody reviewing the case.

That is the property worth having. **A rule that has to be applied by a person protects you only when the person is paying attention,** and book 7 arrived in a batch of twenty-four with nobody reading it.

The character anchor

A heist crew. The job needs a safecracker, a driver, an inside man and a lookout.

SUM is the producer counting talent. Three world-class safecrackers and no driver looks like an outstanding crew on paper, and the total really is high. MIN is the person who has run a job. It asks who is weakest on the team, and if there is nobody in the car the score is zero and the rest of the crew is irrelevant.

Nobody has to add a rule saying the driver matters. It falls out of asking for the minimum.

And it is not a one-off

One rescued false positive is an anecdote, so look at what MIN did across the whole tragedy corpus.

Every Greek tragedy scored $\text{MIN} \geq 0.48$. Every control, epic and didactic text scored $\text{MIN} \leq 0.32$.

Read the shape of that rather than the two numbers. It is not that the tragedies average higher. **There is a gap, and nothing sits inside it.** A difference in averages tolerates any amount of mixing at the boundary. A clean partition does not.

Now the discount, in the same breath. **Nothing was predicted here.** Every text was already known to be a tragedy or a control before the metric ran on it, and separating groups you have already labelled is what episode 1's confirmation table warned about. **MIN was also validated on 10,000-word tragedies only, and Season 3 episode 3 is where that unasked question does its damage.**

The device: what a good instrument looks like when it works

This is worth naming, because the series is mostly failures and the audience should see what success looks like.

a metric chosen for theoretical reasons, before the data	episode 12
a false positive appearing that would have wrecked the results	book 7
the metric rejecting it automatically , with no special case	MIN = 0.00
nobody having to notice	the correction needed no author

The fourth row separates this from luck. The first three describe a narrow escape, where somebody made a defensible choice, a trap appeared, and the choice happened to cover it. **The fourth row is what makes it evidence about the method**, because the correction did not require an author. It would have worked in a text nobody read, on a run nobody watched, in a corpus a hundred times this size.

That is what it looks like when the reasoning was right. Not a number that confirms you. **A number that catches an error you had not spotted yet.**

What would count against this

1. **MIN being right here for the wrong reason.** Book 7 was rejected because its `crimes` bucket is empty, and `crimes` is the rarest bucket in the instrument at **0.95 hits per thousand words**, five times rarer than `violence`. A metric that zeroes out whenever the rarest bucket is empty will zero out **almost everything**, including the texts it is supposed to catch. On that reading MIN detected no false positive at all. It reflected a scarce word-list and got the right answer here by accident. **Season 3 episode 2 settles this, and not in MIN's favour.**
2. **The 0.48 / 0.32 separation being driven by length rather than genre.** The tragedies are of comparable length to each other and the controls are not. **Nobody has run MIN on texts matched for length**, which would separate the two explanations in an afternoon.
3. **One rescued case not being a validation, and it is not.** It is a single text, and it shows the metric can reject something, not that it rejects the right things in general.

How to end the season

The instrument is finished. Written blind, frozen, order checkable, corpus audited, tokens fixed, lexicon de-noised and tagged, metric chosen for the right reason and validated against a false positive it caught by itself.

And nothing has been tested. Not one prediction has been run.

The reasoning that got here is correct. Every decision in thirteen episodes was defensible and most of them were right.

Say that and let it sit, because the whole weight of the next season lands on it. Thirteen episodes have produced an instrument nobody watching has been given a reason to distrust.

Then say the first line of Season 3:

The pre-registered prediction failed, and the controls beat the persecution texts.

Questions

1. Book 7 is the best argument for MIN. **What is the best argument against MIN, and do you have it yet?**
2. MIN was validated on 10,000-word tragedies. **What has not been checked about it?**
3. Book 7 was rejected because a bucket was empty. **Name a second reason a bucket could be empty** that has nothing to do with the scene being peaceful.
4. **Write your prediction for Season 3 now.** Which texts score high, which low, and what fails first?
5. Name a time your own method caught an error **you had not noticed.** If you cannot, what does that tell you about the method?

Runtime

Do not trust a runtime estimate taken from this document's length. Run `audit.py`. It reports the speakable word count per episode with tables, code and question blanks stripped, which is the only number that predicts anything. Divide by 135.

The two-anchor table

CONCEPT	CHARACTER, FOR THEM	CLINICAL, FOR YOU
unfalsifiable	detective who is never wrong, cold reading	100% sensitivity, zero specificity
pre-registration	Ippo announcing the Dempsey Roll, the called shot	pre-specified primary endpoint
the timestamp	sealed envelope with a lawyer	trial registration number
corpus integrity	evidence nobody checked	haemolysed sample
tokenisation	autocorrect betraying you	correct number, wrong units
the sample is part of the claim		post-hoc exclusion criteria
no stemming	autocorrect again	specificity
provenance tags		protocol amendment log
rate per thousand	batting average	prevalence
SUM against MIN	the team that needs every member	N-of-M criteria against all-of

Numbers planted in this season for later detonation

NUMBER	DETONATES IN
crimes at 0.95 per 1000, five times rarer than violence	Season 3, episode 2
a rate is unchanged by shuffling the whole book	Seasons 3 and 4, four times
mythography at 28.65 event-words per 1000	Season 3, the density confound
MIN validated only on 10,000-word tragedies	Season 3, episode 3
round 2's commit contains the scorer	Season 7, the audit

Progress log

DATE	EPISODE	WHAT WENT FUZZY ON CAMERA